

REGAN PETER COSTA, RN

Registered Nurse – Hemodialysis & Nephrology Specialist

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Portfolio & Verified Credentials: zion369369.github.io/regan-peter-costa-portfolio

10+ Years Continuous Dialysis Service (Square Hospitals Ltd.)

BNMC License: 32750 (Valid to July 2029)

BLS Provider (Jan 2025)

Fresenius / B. Braun / SLED Expert

PROFESSIONAL SUMMARY

Dedicated, patient-focused **Registered Hemodialysis Nurse** with **over 10 consecutive years of high-volume clinical practice** in the Hemodialysis & Nephrology Department of **Square Hospitals Ltd.** (Dhaka's leading 400+ bed JCI-standard tertiary private hospital). Possesses proven mastery in the independent setup, priming, programming, and troubleshooting of advanced hemodialysis platforms including **Fresenius 4008S, 5008 CorDiax, B. Braun Dialog+, Gambro AK 96**, and bedside **Sustained Low-Efficiency Dialysis (SLED)** for ICU critical renal care. Accomplished in advanced vascular access cannulation (native Arteriovenous Fistulas and Grafts) using rope-ladder and buttonhole techniques, sterile Central Venous Catheter (**Permcath**) maintenance under strict Aseptic Non-Touch Technique (ANTT), and emergency intradialytic stabilization. Certified BLS provider with active BNMC registration valid through July 2029. Fully prepared for international direct-hire relocation to the UK (NHS direct application), Ireland (HSE), Middle East, or US (EB-3).

CORE CLINICAL COMPETENCIES & TECHNICAL SKILLS

Dialysis Modalities: Intermittent Hemodialysis (IHD), Sustained Low-Efficiency Dialysis (SLED), Online HDF, High-Flux Hemodialysis, Isolated Ultrafiltration (SCUF).	Machinery: Fresenius Medical Care (4008S, 5008 CorDiax), B. Braun Dialog+, Gambro AK 96, Artis Physio, ICU Bedside Dialysis Systems.
Vascular Access: Native AV Fistula (AVF) & Graft (AVG) cannulation (14G–17G), Rope-ladder/Buttonhole, Bruit/Thrill surveillance, Steal syndrome detection.	CVC & Permcath: Tunneled Permcath & Temporary CVC sterile access, Heparin (1000/5000 IU/mL) & Citrate locks, ANTT dressings, Clot clearance protocols.
Emergency Response: Intradialytic hypotension resuscitation, Dialysis Disequilibrium Syndrome (DDS), First-use reaction, Air embolism Durant maneuver, BLS CPR.	Water Plant & Adequacy: Reverse Osmosis (RO) monitoring, Total Chlorine/Chloramine testing (< 0.1 ppm, AAMI/ISO 23500), Urea Kinetic Modeling (Kt/V ≥ 1.3, URR > 68%).
Pharmacotherapy: Blood & blood products transfusion, Erythropoietin (EPO), Darbepoetin, IV Iron sucrose, IV Calcium, Heparinization & Heparin-free saline flush.	Infection Control: Universal precautions, Bloodborne pathogen isolation bays (Hepatitis B/C, HIV), Dialyzer reprocessing safety, Sharps safety.

SPECIALIZED HEMODIALYSIS EXPERIENCE

Senior Staff Nurse – Hemodialysis & Nephrology Unit | Square Hospitals Ltd., Dhaka July 2015 – Present (10+ Years)

- Manage comprehensive specialized hemodialysis care for a high volume of acute and chronic End-Stage Renal Disease (ESRD) and Acute Kidney Injury (AKI) patients in a modern multi-station tertiary unit.
- Independently prime, program, calibrate, and troubleshoot **Fresenius 4008S, 5008 CorDiax, B. Braun Dialog+, and Gambro** systems with zero equipment-related treatment delays.
- Administer bedside **Sustained Low-Efficiency Dialysis (SLED)** for critically ill, hemodynamically fragile renal patients admitted to Medical ICU, Surgical ICU, and Coronary Care Units.
- Perform high-precision cannulation of complex native Arteriovenous Fistulas and Grafts, achieving a >98% first-attempt success rate while minimizing vessel trauma.
- Execute sterile access, flushing, heparin/citrate locking, and dressing changes for temporary and tunneled Central Venous Catheters (**Permcath**) under strict ANTT protocols, maintaining zero CRBSI incidence.
- Swiftly detect and stabilize intradialytic complications, including severe acute hypotension (fluid bolus, UF suspension, Trendelenburg positioning), muscle cramps, and vascular access clotting.
- Administer renal pharmacotherapy including IV Iron sucrose, Erythropoietin (EPO), Darbepoetin, IV multivitamins, and individualized systemic heparinization protocols.
- Perform clinical calculations for dry weight assessment, ultrafiltration (UF) modeling, and adequacy monitoring (Kt/V ≥ 1.3, URR > 68%).
- Conduct daily morning chemical and total chlorine/chloramine tests on the Reverse Osmosis (RO) water purification plant, strictly verifying adherence to AAMI / ISO 23500 water quality standards.
- Verified & Endorsed by Consultant Nephrologist Dr. Mosaddeque Ahmed, MBBS, MRCP (UK), Square Hospitals Ltd.

PRIOR CLINICAL EXPERIENCE

- Senior Staff Nurse – Multi-Specialty Inpatient Wards | Green Life Medical College Hospital

January 2014 – June 2015 (1.5 Years)

 - Delivered clinical bedside nursing care across adult medical and surgical wards, coordinating admissions, preoperative patient prep, postoperative complex wound care, and medication administration.
 - Collaborated within multidisciplinary clinical teams consisting of attending physicians, surgeons, and pharmacists to implement tailored patient care and recovery plans.
 - Responded rapidly to ward emergency calls, participating in emergency code resuscitations and stabilizing acute deteriorating patients.
- Trainee Nurse (Clinical Rotational Internship) | Green Life Medical College Hospital

January 2011 – December 2013 (3 Years)

 - Completed comprehensive supervised rotational internship across Internal Medicine, General Surgery, Pediatrics, Emergency, Obstetrics & Gynecology, and Nephrology units.
 - Maintained accurate, confidential electronic and physical patient records adhering to national clinical documentation standards.

QUALIFICATIONS & PROFESSIONAL CREDENTIALS

Diploma in Nursing Science and Midwifery (3 Years) Green Life Hospital Nursing Institute Bangladesh Nursing Council (BNC) Session: 2010–2011 Comprehensive Council Exam Passed: April 2014	Registered Nurse & Midwife Practitioner (BNMC) Bangladesh Nursing & Midwifery Council Registration No: 32750 Original Reg: 07-07-2014 Renewed: 06-07-2024 Valid Upto: 05-07-2029
Basic Life Support (BLS) Provider Certification Approved by Bangladesh Society of Emergency Medicine (BSEM) Training Center: Dr. Nizam Medical Centre Certified: 28 January 2025	Secondary & Higher Secondary Education H.S.C: Tejgaon College, Dhaka Board (Business Studies, 2010) S.S.C: Tumilia Boys High School, Dhaka Board (Business Studies, 2008)

CLINICAL EQUIPMENT & TECHNICAL PROFICIENCY LOG

Fresenius 4008S / 5008:	Fully independent operation, dialyzer circuit priming, blood line setup, profiling (Na/UF), disinfection, pressure alarm troubleshooting.
B. Braun Dialog+:	Independent therapy setup, cartridge bloodline mounting, bio-filtration, bicarbonate preparation, and routine maintenance disinfection.
Gambro AK 96:	Setup, arterial/venous chamber level adjustment, isolated ultrafiltration (SCUF), treatment data logging, and machine descaling.
ICU Bedside SLED:	Bedside slow-flow dialysis in ICU/CCU, continuous heparinization, pre/post-filter monitoring, hemodynamic safety stabilization.
Vascular Cannulation:	Native radiocephalic & brachiocephalic AV Fistulas, PTFE Arteriovenous Grafts, rope-ladder & buttonhole puncture (14G–17G needles).
Permcath & CVC Care:	Tunneled Internal Jugular Permcath access, exit-site ANTT dressing, 5000 IU/mL heparin locks, 4% citrate instillation, clot aspiration.

LANGUAGES, IT SYSTEMS & DECLARATION

Languages:	English: Full Professional Working Proficiency (clinical handovers, patient documentation, medical terminology). Bengali: Native / Mother Tongue.
Digital Systems:	Hospital Information Systems (HIS), Electronic Medical Records (EMR), Microsoft Office (Word, Excel, PowerPoint).
Declaration & References:	I hereby declare that all clinical records, employment history, and credentials stated herein are authentic. Verified reference letters, original council certificates, and clinical competency logs available immediately upon request.